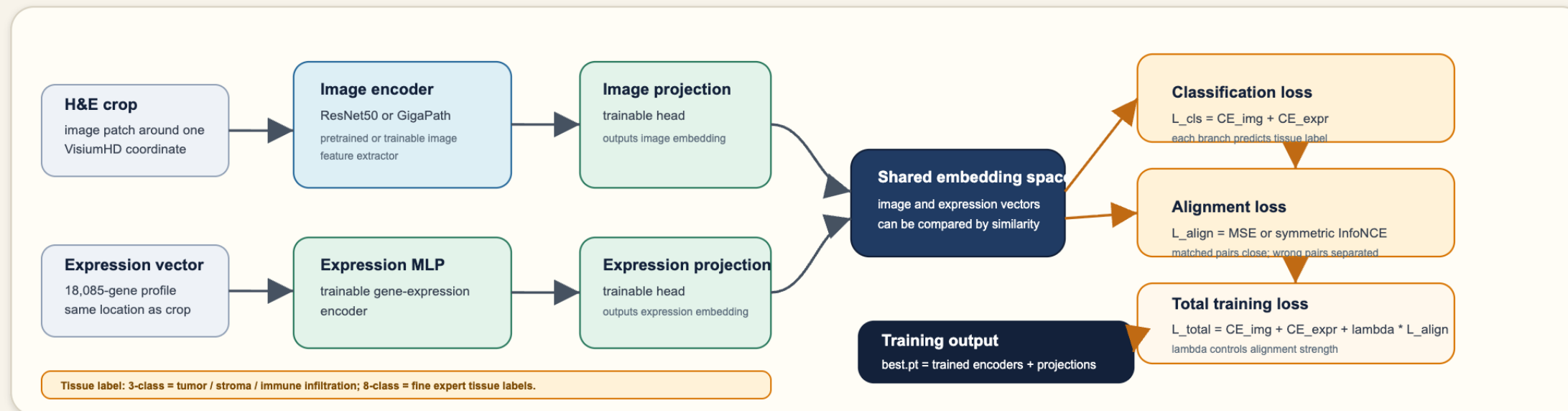


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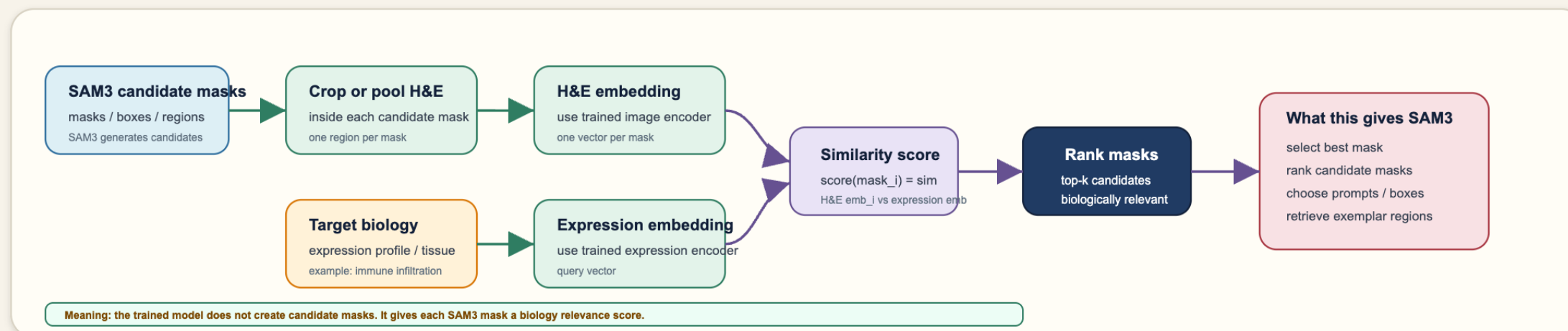
From H&E-Expression Training to SAM3 Mask Ranking

The training does not output masks. It learns a shared embedding model that later scores SAM3 candidate masks by biological relevance.

1. Training stage: learn a shared H&E-expression embedding model



2. Use after SAM3 has candidate masks: score and rank masks by target biology



One sentence: SAM3 cuts possible regions; the H&E-expression model decides which regions best match the target biology.

Training Setup

- **Optimizer:** AdamW
- **Learning rate:** $1e-4$ for expression encoder, projection heads, and classifiers
- **Dropout:** 0.3
- **Training length:** up to 8 epochs
- **Early stopping:** patience = 2
- **Monitor:** validation expression macro-F1
- **Class imbalance:** balanced sampler + sqrt inverse class weights

Encoder Strategy

- **Frozen baseline:** GigaPath frozen
- **Fine-tuning tests:** frozen, last-1-block, last-2-block
- **LR schedule tests:** constant LR vs warmup + cosine schedule

Loss Setup

- **Main loss:** $L_{total} = CE_{img} + CE_{expr} + \lambda L_{align}$
- **CE_{img} :** image embedding predicts tissue label
- **CE_{expr} :** expression embedding predicts tissue label
- **Alignment loss:** MSE or symmetric InfoNCE
- **InfoNCE λ sweep:** $0.05, 0.10, 0.20, 0.50$

Tests Performed

1. **Backbone baseline**
ResNet50 vs GigaPath
Purpose: choose the image encoder for H&E morphology.
2. **Crop size and label granularity**
crop64 vs crop128; 3-class vs 8-class
Purpose: choose stable input context and tissue-label resolution.
3. **Loss ablation**
CE-only, CE + MSE, CE + InfoNCE, InfoNCE-only
Purpose: separate tissue classification from cross-modal alignment.
4. **InfoNCE weight sweep**
 $\lambda = 0.05, 0.10, 0.20, 0.50$
Purpose: tune alignment strength.
5. **GigaPath training strategy**
frozen vs last-1-block vs last-2-block; constant LR vs warmup + cosine
Purpose: test whether partial fine-tuning improves the foundation encoder.
6. **Retrieval evaluation**
expression-to-image Recall@1, Recall@5, Recall@10; fused R@1
Purpose: test whether expression queries can retrieve matching H&E regions.
7. **SAM3 mask-ranking use case**
use learned similarity to rank SAM3 candidate masks
Purpose: connect the trained embedding model to the final segmentation workflow.

1. Backbone Baseline: ResNet50 vs GigaPath

Metric	ResNet50	GigaPath	Interpretation
3-class expr macro-F1	0.6830	0.6862	Expression branch is similar
3-class image acc	0.6212	0.7488	GigaPath gives better H&E morphology features
3-class fused R@1	0.7605	0.8606	GigaPath gives better biology-guided retrieval
8-class image acc	0.4284	0.5892	GigaPath is stronger on harder fine labels
Best fine-tuned image acc	lower / not main	0.8312	Last-2-block fine-tuning further improves GigaPath

Conclusion: GigaPath is the better main backbone. It only slightly improves expression classification, but it clearly improves H&E image accuracy and fused retrieval, which are the most relevant parts for ranking SAM3 candidate masks.

2. Crop / Label Granularity: crop size and 3-class vs 8-class

Comparison	Easier setting	Harder setting	Result
Crop size	crop64	crop128	crop128 became the main setting because it keeps more tissue context
Label granularity	3-class	8-class	8-class is harder and less stable
GigaPath expr macro-F1	0.6862	0.4555	fine labels reduce expression classification performance
GigaPath image acc	0.7488	0.5892	fine labels are harder for H&E morphology
GigaPath fused R@1	0.8606	0.8439	retrieval remains strong, but 3-class is more stable
8-class + fine-tuning	frozen: 0.5892 image acc	last-2 fine-tune: 0.7084 image acc	fine-tuning helps the harder 8-class setting

Conclusion: 3-class crop128 is the most stable main setting. 8-class is biologically more detailed, but harder because the labels are finer, noisier, and more imbalanced.

3. Loss Ablation: What Loss Actually Helps?

Loss setting	Best expr macro-F1	Best image acc	Best fused R@1	Interpretation
CE only	0.6830	0.7336	0.8558	strong classification baseline
CE + MSE	0.6724	0.7340	not main retrieval	original alignment baseline
CE + InfoNCE	0.6862	0.7488	0.8606	best balance of classification + retrieval
InfoNCE only	0.2647	0.2812	0.8541	retrieval survives, but classification collapses
Best λ	$\lambda = 0.10$	$\lambda = 0.50$	$\lambda = 0.50$	optimal λ depends on metric

Conclusion: CE supervision is necessary for stable tissue classification. InfoNCE improves cross-modal retrieval, but InfoNCE alone is not enough. The best main direction is **CE_img + CE_expr + λ InfoNCE**.

4. Training Strategy: Frozen vs Fine-Tuned GigaPath

Strategy	Expr macro-F1	Image acc	Fused R@1	Interpretation
Frozen + constant LR	0.6799	0.6912	0.8566	stable baseline
Frozen + warmup/cosine	0.6796	0.6908	0.8567	scheduler alone changes little
Last-1-block fine-tune	0.6929	0.7920	0.8740	improves image branch and retrieval
Last-2-block fine-tune	0.6945	0.8312	0.8804	best overall strategy

Purpose: test whether GigaPath should stay frozen or be adapted to this VisiumHD task.

Conclusion: partial fine-tuning helps. Last-2-block fine-tuning gives the best image accuracy and fused retrieval, so it is the strongest current training strategy.

5. Retrieval Evaluation: Can Expression Find Matching H&E Regions?

Setting	Best R@1	Best R@5	Best R@10	Interpretation
GigaPath 3-class	0.7450	0.8693	0.9109	strong expression-to-image matching
GigaPath 3-class fused	0.8804	—	—	best SAM3-relevant ranking signal
GigaPath 8-class	0.6128	0.7368	0.7766	harder but still useful
ResNet50 3-class	0.7286	0.8907	0.9422	good retrieval, weaker fused ranking
ResNet50 8-class	0.5531	0.7986	0.8730	retrieval works, but image branch weaker

Purpose: test whether the learned shared space can use an expression query to retrieve matching H&E regions.

Conclusion: retrieval works well, especially for GigaPath 3-class. This supports the SAM3 use case: expression / target biology can rank candidate masks by biological relevance.

6. Overall Takeaways and Next Step

Finding	Evidence	Meaning
GigaPath is the best backbone	higher image acc and fused R@1 than ResNet50	better H&E morphology representation
crop128 + 3-class is most stable	higher macro-F1 and image acc than 8-class	good main baseline
8-class is harder but useful	lower macro-F1 / image acc, improved by fine-tuning	fine labels need stronger training
CE supervision is necessary	InfoNCE-only classification collapses	labels give tissue structure
InfoNCE improves alignment	CE + InfoNCE gives best balance	better for retrieval-style SAM3 ranking
Last-2-block fine-tuning is strongest	best image acc and fused R@1	current best checkpoint direction

Conclusion: The strongest current direction is **GigaPath crop128 3-class + CE_img + CE_expr + InfoNCE + last-2-block fine-tuning**.

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